

Frontal Theta Phase Synchrony and Post-Error Slowing: A Psychobiological and Psychometric Investigation of Error Monitoring in a Large Open EEG Dataset

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Colour key and document status

Black: original text or new passages explicitly labelled “Proposed replacement wording”. Red: replacement wording or additions addressing supervisor feedback. Green: choices or factual details requiring confirmation. This is a re-typeset review copy, not the original submitted file or a reproduction of its pagination. The submitted word count of 2,144 does not apply to this annotated version.

The supervisor requested greater specificity; they did not select the exact tests, wavelet settings, task aggregation or security arrangements below. Red methodological wording is a concrete recommendation for review, not evidence of supervisor approval or implementation. Green notes identify what must be settled before the document is treated as a complete protocol.

Suggested correction — This revised protocol provides explicit methodological recommendations in response to supervisor feedback. The researcher will review and record the selected settings before confirmatory testing. Unconfirmed governance details will be completed from actual arrangements and programme requirements.

Proposal Declaration — reproduced from submission

I, Moyosore Olasupo, declare that this proposal has been composed solely by me and has not been submitted, in whole or in part, in any previous application for a degree. Except where stated otherwise, by reference or acknowledgement, the work presented is entirely my own and has been generated by me as a result of my own original research. I acknowledge using Open AI’s GPT-6 Astra to support my understanding of concepts, provide feedback, and assist with planning. I have not plagiarised AI generated content and I have reviewed the materials included in this proposal to ensure that it reflects my own understanding. Signed: Date: 13/09/2026

The declaration above is reproduced as part of the original document. It has not been newly signed. Any declaration for resubmission must accurately describe assistance with that version and follow university requirements.

Suggested correction — I take responsibility for the content of this revised proposal and for checking its sources, methods and conclusions. Assistance was used for understanding, planning, drafting and revision, proofreading, reference preparation and preliminary code development. This wording must be checked against the required university declaration before

signing.

Background Literature Review

Successful goal-directed behaviour requires individuals to monitor their actions and adjust their responses when a mistake occurs. However, detecting an error does not necessarily improve subsequent performance (Danielmeier & Ullsperger, 2011). The study of how neural error signals are related to behavioural adjustment is important for understanding cognitive control. My literature review search was conducted by using APA PsycINFO, covering error monitoring, frontal theta activity, post-error slowing (PES), and reliability measurement. Error-related negativity (ERN) is an event-related potential that occurs shortly after an incorrect response and is measured at frontocentral scalp electrodes, including during flanker tasks (Holroyd & Coles, 2002). It provides an established measure of error processing. Theta power describes the strength of activity within a frequency band and is related to squared amplitude. Inter-site phase synchrony describes the consistency of phase relationships between signals recorded at different electrodes. These measures capture different aspects of the neural response to errors.

Prior Findings

Cavanagh et al. (2009) found increased theta synchrony between midfrontal and lateral-frontal EEG sites during error trials in a flanker task. Greater synchrony was associated with subsequent post-error slowing (PES), suggesting a relationship between synchronised frontal activity and behavioural adjustment. Valadez and Simons (2018) additionally found that error-related frontal theta power and PES predicted performance recovery. However, findings concerning theta power do not directly establish the behavioural relevance of connectivity, and these associations alone do not demonstrate causation. Danielmeier and Ullsperger (2011) reviewed evidence that PES and improved post-error accuracy are not consistently associated. This means that a slower response does not automatically indicate successful adaptation. Murphy et al. (2016) found that larger error-evoked pupil responses predicted greater slowing and a higher probability of a correct subsequent response in a task with long inter-stimulus intervals. This suggests that the timing of subsequent task demands influences whether an error response supports beneficial adjustment. Volume conduction poses a challenge for interpreting scalp connectivity. Vinck et al. (2011) introduced the weighted phase-lag index (wPLI) to estimate phase synchrony with reduced sensitivity to volume conduction effects and noise. They also developed a debiased estimator of its squared value to reduce sample-size bias. This provides a basis for estimating lagged connectivity. Reliability must be assessed when interpreting individual differences. Clayson et al. (2021) distinguished measurement precision from internal consistency, explaining why good data quality does not automatically establish reliable measurement of differences between individuals. Consequently, the reliability of error-related theta synchrony should be assessed alongside its relation with behaviour.

Theory

The reinforcement-learning account proposes that the ERN reflects a negative learning signal carried through the dopamine system to the anterior cingulate cortex, contributing to subsequent performance adjustment (Holroyd & Coles, 2002). This provides a connection between error monitoring and behavioural adaptation. A complementary network interpretation proposes that theta synchrony supports coordination between frontal regions involved in performance monitoring and implementing adjustments. The findings of Cavanagh et al. (2009) are consistent with this interpretation, but do not establish direct communication between particular anatomical regions. A positive association between theta connectivity and PES would support a neural-behavioural relationship. Examining this relationship separately from measurement reliability allows the study to address both the behavioural relevance and psychometric properties of frontal theta connectivity.

Rationale and Hypotheses

Previous findings provide a basis for examining whether error-related frontal theta synchrony is associated with PES (Cavanagh et al., 2009). This study will investigate the relationship between participants using the ds004883 flanker dataset. A secondary comparison will test whether connectivity is more reliable than ΔERN ; this advantage has not yet been established. This study will use the publicly available EEG dataset ds004883. Its retained behavioural information supports trial reconstruction, although final eligibility for the planned analyses will depend on the specified screening criteria. The research questions are: Is error-related frontal theta phase synchrony associated with post-error slowing (PES) between participants? Does error-trial theta synchrony demonstrate higher internal consistency than ΔERN at matched numbers of usable error trials?

H1: Error responses will produce a more negative mean response-locked ERP amplitude at FCz than correct responses during 0–100 ms after the response. Prediction: $\Delta\text{ERN} = \text{ERN} - \text{CRN} < 0$.

H2: Frontal theta synchrony, measured using dWPLI between FCz and lateral-frontal electrodes, will be greater on error trials than correct trials.

H3: Individuals with stronger error-related theta synchrony will show greater PES. $\text{PES} = \text{RT}(E+1) - \text{RT}(E-1)$, using valid correct responses immediately surrounding an error.

H4a: Error-trial theta dWPLI will show higher Spearman–Brown-corrected split-half reliability than ΔERN at matched numbers of usable error trials.

H4b: Examine the stability of the H4a reliability comparison across repeated random splits of the same trials. This is a robustness analysis, not a separate formal hypothesis.

Aims and Objectives

The primary aim is to examine the relationship between error-related frontal theta synchrony and PES, and the secondary aim is to compare connectivity and ΔERN reliability. The objectives are to (i) evaluate error-related EEG effects, (ii) test the between-participant association between synchrony and PES, and (iii) compare reliability using matched error-trial counts.

Research Design

Design

The study will use a quantitative secondary analysis of EEG and behavioural data from the ds004883 flanker dataset (Clayson et al., 2024). It will combine within-participant comparisons of error and correct responses with a between-participant analysis of frontal theta synchrony and post-error slowing (PES). No additional participants will be recruited. H1 will compare mean response-locked ERP amplitudes between error and correct trials at FCz. H2 will compare theta synchrony between the response conditions. H3 will examine whether participants with stronger error-related connectivity have higher PES. H4a will compare the Spearman-Brown-corrected split-half reliability of connectivity and Δ ERN using matched numbers of usable error trials. H4b will assess stability across repeated random splits of the same trials.

Materials

The study will use EEG recordings and associated behavioural metadata from the ds004883 flanker dataset (Clayson et al., 2024). The original study presented three arrow-flanker task variants using E-Prime 3.0. EEG was recorded using an Electrical Geodesics, Inc. HydroCel Geodesic Sensor Net with 128 scalp sites and a vertex reference, sampled at 500 Hz. The dataset contains EEGLAB SET/FDT files, electrode information and event metadata. Behavioural fields retained in the SET files include response times, response accuracy and original trial identifiers. These fields will provide information for reconstructing behavioural sequences. EEG processing and analysis will use MNE-Python 1.12.1, supported by NumPy, SciPy, and Matplotlib. Picard will be used for independent component analysis. The standard montage provides approximate correspondences between E6 and FCz, E27 and F5, and E123 and F6. These represent standard electrode locations rather than individual anatomical localisation.

Participants

The source dataset contains records for 172 participants recruited from undergraduate courses at Brigham Young University and the University of South Florida. According to Clayson et al. (2024), participants received course credit, and the eligibility criterion specified an age over 18 years. Eligibility excluded individuals reporting psychiatric or neurological diagnoses, psychoactive medication use, uncorrected visual impairment or limited English fluency. All participants whose data meet the prespecified, analysis-specific eligibility criteria will be considered for the secondary analysis. Inclusion will depend on EEG quality and sufficient usable error and correct trials. The synchrony PES analysis will additionally require sufficient valid behavioural sequences. Consequently, eligible sample sizes may differ across the ERP, connectivity, behavioural-association and reliability analyses. Participants will be included according to predefined data-quality and trial-count criteria. The final sample size and reasons for exclusion will be reported for each analysis. Statistical sensitivity will be evaluated using the number of eligible participants.

Data Analysis Plan

Before hypothesis testing, response-time exclusions, minimum usable trial counts, task-variant handling, connectivity parameters and statistical decision rules will be specified. EEG preprocessing will include filtering, referencing, ocular-artefact correction and response-locked epoching, with documented quality checks and exclusions. Original trial identifiers will be preserved when linking EEG epochs to behavioural sequences. Recordings from the same participant will not be treated as independent observations.

Before confirmatory testing, record the exact RT rules, minimum usable trial and triplet counts, and treatment of the three task variants. The original proposal does not settle these. The paired tests below require one matched error/correct score per participant; the rule for producing that score across variants must be agreed first. The complete-variant averaging and numerical thresholds in the longer working draft remain recommendations, not established requirements.

Suggested correction — Working protocol recommendation: calculate each measure separately for Tasks A, B and C, then average the three task scores equally for each eligible participant. Require at least 10 usable error and 10 usable correct epochs per variant for H1/H2; H3 additionally requires at least 10 eligible correct-error-correct triplets per variant with retained central error epochs. Exclude missing, non-finite and non-positive RTs at any triplet position, ambiguous responses, omissions and crossings of uncertain block boundaries. Retain original trial positions and remove only explicitly identified practice trials. Do not impose an undocumented physical-onset correction or sample-dependent upper RT trimming. These are pragmatic recommended rules, not validated sufficiency thresholds; assess feasibility before adopting them and report their limitations.

For H1, ERN and CRN will be calculated as mean response-locked ERP amplitude at FCz/E6 during 0–100 ms after error and correct responses. $\Delta\text{ERN} = \text{ERN} - \text{CRN}$, and response-locked waveforms will be plotted. A two-sided paired-samples t-test will compare participant-level error and correct scores, equivalently testing whether the mean participant-level ΔERN differs from zero. Support for the predicted direction requires a negative difference and a significant multiplicity-adjusted result. Report the mean difference in microvolts, its 95% confidence interval, t statistic, degrees of freedom, adjusted p-value and paired effect size d_z .

Proposed replacement wording — For H1, mean response-locked amplitudes at FCz/E6 will be calculated over 0–100 ms for error and correct trials. A two-sided paired-samples t-test will compare the participant-level condition scores. The predicted difference is $\Delta\text{ERN} = \text{ERN} - \text{CRN} < 0$. Results will include the mean difference, its 95% confidence interval, t statistic, degrees of freedom, raw and multiplicity-adjusted p-values, and paired effect size d_z .

The H1 and H2 paired tests treat participants as independent units and assess within-participant differences. Exact small-sample t inference assumes normally distributed differences. Inspect difference distributions and influential observations as diagnostics; do not select a different test according to which produces significance. Report assumption concerns and label additional robust analyses as sensitivity analyses. The implementation is documented by the SciPy community (n.d.).

Proposed replacement wording — The H1 and H2 analyses will use participants as independent units and retain the pairing of error and correct scores. Participant-level differences will be inspected for distributional problems and influential values. Assumption concerns will be reported, and additional robust analyses will be identified as sensitivity analyses rather than

selected according to statistical significance. Paired tests will use the procedure documented by the SciPy community (n.d.).

For H2, use the debiased estimator of squared weighted phase-lag index (Vinck et al., 2011), called dWPLI here, for E6-E27 and E6-E123. Estimate connectivity across trials separately at each time and frequency before averaging across the scoring window and frequency centres. Average the two pair scores with equal weights; do not select the pair showing the largest effect or pool pair-specific cross-spectra before estimation. Negative finite-sample estimates will not be truncated and a square root will not be applied.

Proposed replacement wording — For H2, frontal theta connectivity will be estimated using the debiased estimator of squared wPLI (Vinck et al., 2011). Estimates will be calculated across trials separately for E6-E27 and E6-E123 at each time-frequency point, averaged across the specified scoring samples and frequencies, and then averaged equally across the two electrode pairs. Negative estimates will be retained and no square-root transformation will be applied.

Proposed time-frequency settings: zero-mean complex Morlet wavelets at 4, 5, 6, 7 and 8 Hz, three cycles at each frequency, estimated on -1.0 to $+1.15$ s response-locked epochs before scoring 0–150 ms. These values require agreement and implementation checks. They are project choices, not settings prescribed by the supervisor. Three cycles broaden spectral sensitivity; 4–8 Hz denotes centre frequencies, not a sharply bounded passband.

Suggested correction — Working protocol recommendation: use zero-mean complex Morlet wavelets at 4, 5, 6, 7 and 8 Hz with three cycles per frequency. Transform clean response-locked epochs from -1.0 to $+1.15$ s before selecting 0–150 ms. Estimate debiased squared wPLI across trials at each time-frequency point, average equally across the selected samples and frequencies, and then average E6-E27 and E6-E123 equally. Verify full wavelet support and discontinuity handling before execution. The short cycle count imposes broad spectral sensitivity and temporal smoothing.

Report wavelet duration as well as cycles, following Cohen (2019). Under the proposed three-cycle implementation, the amplitude-envelope full width at half maximum is approximately 281 ms at 4 Hz and 141 ms at 8 Hz. MNE's numerical wavelet support extends approximately five Gaussian standard deviations in either direction (MNE-Python contributors, n.d.). Full signal support must be present without crossing recording discontinuities. A 0–150 ms scoring window therefore does not isolate activity strictly within the first 150 ms after the response.

Proposed replacement wording — Wavelet cycles and amplitude-envelope duration will be reported following Cohen (2019). For the proposed three-cycle wavelets, temporal full width at half maximum is approximately 281 ms at 4 Hz and 141 ms at 8 Hz. Epochs will provide the complete numerical wavelet support without crossing recording discontinuities (MNE-Python contributors, n.d.). The 0–150 ms connectivity score will therefore be interpreted as temporally smoothed activity rather than an effect confined to that interval.

Within each participant and variant, match error and correct estimates to the smaller usable condition count. Sample without replacement, repeat 1,000 times with seed 42, and average the estimates to obtain one score per condition. These repeated draws are not independent observations. Compare participant-level error and correct connectivity using a two-sided paired-samples t-test. Report the mean difference, 95% confidence interval, d_z , t statistic, degrees of freedom and adjusted p-value. The predicted error-minus-correct difference is positive.

Proposed replacement wording — Within each participant and task variant, error and correct trials will be matched to the smaller usable condition count through sampling without replacement. Connectivity estimates will be averaged across 1,000 draws using random seed 42. The resulting participant-level condition scores will be compared using a two-sided paired-samples t-test, with a positive error-minus-correct difference predicted. The mean difference, 95% confidence interval, d_z , t statistic, degrees of freedom, and raw and adjusted p-values will be reported. Repeated draws will not be treated as independent observations.

For H3, post-error slowing will be calculated as $PES = RT(E+1) - RT(E-1)$, using valid correct responses immediately before and after an error (Dutilh et al., 2012). Triplets will be screened for omissions, incorrect flanking responses, response-time exclusions and crossings of task-block boundaries or recording discontinuities. Neighbours will be identified from the original trial sequence before exclusions are applied. The association between participant-level error-trial synchrony and mean valid PES will be assessed using Spearman's correlation.

H3 will still be tested if H2 is unsupported. In that case, an association would link connectivity measured during error trials with PES, but would not establish that this connectivity is specifically enhanced by errors. A non-significant H2 result is not evidence of equality; its effect size and interval will inform interpretation. H3 cannot retrospectively establish support for H2.

Proposed replacement wording — H3 will be evaluated regardless of the H2 result. If H2 is unsupported, a significant H3 association will be interpreted as a relationship between connectivity measured on error trials and PES, without establishing that errors specifically enhance that connectivity. The H2 effect estimate and confidence interval will inform this interpretation; a non-significant result will not be treated as evidence that the response conditions are equivalent.

For H4a, Spearman-Brown-corrected split-half reliability will be compared between error-trial synchrony and ΔERN using matched usable error trials and the same eligible participants within each comparison. Correct-trial sampling for ΔERN will be predefined. Reliability will be examined across a prespecified range of usable error-trial counts, informed by dataset feasibility and relevant methodological evidence. Each comparison will use matched trial counts for dWPLI and ΔERN , with the number of trials contributing to each half stated explicitly.

Confirm the reliability trial-count grid, total versus per-half counts, correct-trial allocation and inferential procedure before testing H4a. The longer working revision proposes explicit sampling and participant-bootstrap interval rules; these require review, especially their feasibility and coverage for the transformed reliability difference. Counts or thresholds must not be attributed to the supervisor.

Suggested correction — Working protocol recommendation: examine $N = 20, 40, 60, \dots$ total usable error trials per variant, with $N/2$ errors in each half. Retain counts supported by at least 30 participants with N eligible error and N eligible correct trials in every variant. Freeze this grid using quality-control counts before estimating reliability. Use identical sampled errors and half assignments for both measures, and N independently sampled correct trials split equally for ΔERN . Compute Pearson half-score correlations across the same participants and apply $r_{SB} = 2r/(1+r)$. Compare corrected reliabilities using 20,000 paired participant bootstrap resamples, keeping each participant's four half scores together and the primary trial partition fixed. Report undefined or unstable estimates rather than silently discarding them. These count floors are proposed feasibility rules, not established adequacy guarantees.

For H4b, repeated random split-half partitions will assess how sensitive the reliability comparison is to the division of trials. Results will be summarised descriptively. Because the partitions reuse the same participants and trials, their reliability coefficients will not be treated as independent observations.

For H1-H3, adjust the three two-sided p-values together using Holm's method at familywise $\alpha = .05$ (Holm, 1979). H3 remains the primary scientific question. Report raw and adjusted p-values and interpret support only in the predicted direction. Conventional 95% intervals will be identified as unadjusted intervals. H4b remains descriptive and creates no additional hypothesis tests.

Proposed replacement wording — The two-sided H1, H2 and H3 tests will form one confirmatory family, with Holm correction at familywise $\alpha = .05$ (Holm, 1979). H3 remains the primary research question. Raw and adjusted p-values will be reported, and support for each directional prediction will require a significant adjusted result in the predicted direction. Accompanying 95% intervals will be labelled as unadjusted. H4b will remain descriptive.

H4a needs its own explicitly defined secondary multiplicity rule once the number and form of reliability comparisons are agreed. The working revision recommends Bonferroni-adjusted participant-bootstrap intervals across feasible trial counts. That is approximate familywise control conditional on the selected trial partitions; it is not proof of exact coverage or project-wide .05 error control across both families.

Suggested correction — Working protocol recommendation: if K reliability trial counts are eligible, use the $\alpha/(2K)$ and $1-\alpha/(2K)$ percentile endpoints of the paired participant-bootstrap distribution, with $\alpha = .05$. An adjusted interval wholly above zero supports higher connectivity reliability at that count; an interval including zero does not establish a difference. This provides approximate Bonferroni-adjusted interval inference within the secondary H4a family, conditional on the primary partition. Report instability and keep H4b partition distributions descriptive. H1-H3 retain their separate Holm-adjusted family.

Report effect sizes, uncertainty intervals, usable trial counts, participant exclusions and analysis-specific sample sizes. Maintain a dated log of all preprocessing, screening and modelling decisions, stating what information had already been inspected. Existing behavioural audits and single-recording EEG checks preceded this revision. Choices informed by those audits will be described as data-informed feasibility decisions, not decisions made before any data inspection. Later changes prompted by hypothesis results will be labelled exploratory.

Proposed replacement wording — Effect sizes, uncertainty intervals, usable trial counts, exclusions and analysis-specific sample sizes will be reported. A dated decision log will record each preprocessing, screening or modelling choice and the information available when it was made. Decisions informed by the existing behavioural audits and single-recording EEG checks will be identified as data-informed. Subsequent changes prompted by hypothesis results will be reported as exploratory.

Procedure

The study will use existing EEG and behavioural data from the ds004883 dataset. Trial-level behavioural information will be assessed to establish whether reaction times, response accuracy and trial sequence support the calculation of a meaningful post-error behavioural outcome. EEG and behavioural measures will be derived using the specified analysis procedures. The primary analysis will examine the relationship between error-related frontal theta synchrony and post-error slowing. A secondary analysis will compare the Spearman-Brown-corrected split-half reliability of theta synchrony and Δ ERN using matched numbers of usable error trials. Repeated random splits will assess the stability of this comparison without treating the resulting coefficients as independent observations. Analysis-specific exclusions, usable trial counts and participant numbers will be documented.

Ethics

The study will use existing de-identified EEG and behavioural data from the publicly available ds004883 dataset. Clayson et al. (2024) report that participants provided written informed consent covering public data sharing and its associated risks. No participants will be contacted and no new data will be collected. The project will follow the University of Essex Online's Research Ethical Approval Policy and programme-specific guidance (University of Essex Online, 2025). The required application documents and review arrangements will be confirmed with the module tutor.

No attempt will be made to identify participants or link their records to identifying information. Original data will remain unchanged, with derived outputs stored separately. The governance plan covers working storage, access, backup, retention, sharing and disposal, including locally generated participant-level files.

Proposed replacement wording — No attempt will be made to identify participants or connect dataset records with identifying information. Source files will be preserved unchanged and derived outputs stored separately. The data-governance plan will specify storage, authorised access, backup, retention, sharing and disposal arrangements for both source data and locally generated participant-level files.

Confirmed project locations: source files are under data/ds004883_Clayson/; derived audits and results under results/; figures under figures/; analysis code under scripts/ and notebooks/, within /home/moyo/neuro_projects/essex_project/. These locations establish separation of source and derived material, but do not establish encryption, exclusive access, institutional approval or a working backup.

Proposed replacement wording — The working project is located at /home/moyo/neuro_projects/essex_project/. Source data are stored in data/ds004883_Clayson/, derived audits and results in results/, figures in figures/, and analysis code in scripts/ and notebooks/. Storage protection, authorised access, backups and institutional approval will be documented separately using confirmed arrangements.

Host and ownership: confirm whether the research host is a personal computer or university/shared server, who owns/administers it, and whether it is approved for this work.

Suggested correction — The working dataset is stored on [HOST/DEVICE NAME AND TYPE], owned by [OWNER] and administered by [ADMINISTRATOR], under /home/moyo/neuro_projects/essex_project/. Source data remain in data/ds004883_Clayson/ and

derived participant-level outputs in results/. This location is [APPROVAL STATUS AND BASIS]. Complete the bracketed fields from verified facts before use.

Access: name the researcher and any other account holders or administrators able to access the files. State how supervisor access will be provided. Do not describe access as researcher-only unless verified.

Suggested correction — Access to research files is available to [NAMES OR ROLES], including [HOST ADMINISTRATORS, IF APPLICABLE], through [AUTHENTICATION AND PERMISSIONS]. Supervisor access will use [APPROVED TRANSFER OR ACCESS ROUTE]. Access will be reviewed when a person's role ends. Complete these fields from actual permissions; do not claim exclusive access without verification.

Security and backup: confirm the actual account protections, storage/transfer encryption, backup destination, access to backups, backup frequency and restore-check arrangements. The proposed plan is authenticated restricted access, encrypted storage/transfers and a separate protected backup; these are not verified existing safeguards.

Suggested correction — Working files use [ACTUAL STORAGE ENCRYPTION AND ACCOUNT PROTECTIONS]. Transfers use [SECURE TRANSFER METHOD]. Backups are kept at [DESTINATION], accessible to [AUTHORISED PEOPLE], protected by [BACKUP SECURITY], and made [FREQUENCY]. Restore checks occur [SCHEDULE]. If a control or backup is absent, record that accurately and specify the implementation action before presenting it as established.

Retention: obtain the programme's applicable retention period, the event from which it runs, and the responsible person. Record how it applies to downloaded data, processed epochs, participant-level tables, code and the final report. No retention duration has been confirmed.

Suggested correction — Under [APPLICABLE PROGRAMME/POLICY AND VERSION], [RESPONSIBLE PERSON] will retain [SPECIFIED SOURCE, DERIVED AND PROJECT RECORDS] for [PERIOD], measured from [START EVENT]. The resulting review or disposal date is [DATE]. Any different rules for code, the final report or participant-level files will be recorded. Obtain the programme's confirmed requirement before completing this statement.

Sharing and disposal: confirm permitted sharing routes for code, aggregate results and any participant-level derivatives, and the deletion/backup-expiry procedure at the retention endpoint. Review notebook outputs before sharing; public source availability does not automatically authorise every new sharing arrangement.

Suggested correction — Code and aggregate results will be shared through [APPROVED ROUTE] after reviewing outputs for participant-level information and applicable conditions. Participant-level derivatives will be [PERMITTED SHARING OR RESTRICTED ACCESS ARRANGEMENT]. At the retention endpoint, [RESPONSIBLE PERSON] will apply [DELETION METHOD], including [BACKUP EXPIRY/REMOVAL PROCEDURE], and record disposal. These fields require confirmed arrangements.

Institutional review: confirm the required application route and current approval status with the module tutor. This colour copy does not establish ethical approval.

Suggested correction — The project follows [CONFIRMED PROGRAMME REVIEW ROUTE]. Its current status is [NOT SUBMITTED / UNDER REVIEW / APPROVED, AS APPLICABLE]. Where approval has been granted, record [REFERENCE, DATE AND SCOPE]. Further work will follow

the tutor’s written guidance and any approval conditions. Do not substitute “approved” without documentary evidence.

Limitations

This study uses an existing dataset, so the available task design, recordings and behavioural information cannot be modified. Differences between the three flanker variants may influence EEG responses and post-error behaviour, requiring explicit consideration in the analysis. Participant eligibility will depend on EEG quality, usable error trials and valid behavioural sequences. Exclusions may reduce the analysable sample and limit generalisability. Small error-trial counts may also produce imprecise synchrony and reliability estimates, particularly when trials are divided into halves. Post-error slowing is a behavioural index rather than a direct measure of adaptive cognitive control. Comparing responses immediately before and after errors addresses some local fluctuations in response speed but does not eliminate all influences, including trial difficulty and congruency. Uncertainty about the correspondence between recorded event markers and physical stimulus timing also limits interpretation of absolute reaction times. Scalp EEG provides limited anatomical specificity. Although dWPLI reduces sensitivity to some zero-lag effects and sample size bias, it does not eliminate artefact contamination or establish direct communication between brain regions. Standard electrode correspondences do not represent individual anatomical localisation, and time frequency smoothing limits the temporal specificity of connectivity estimates. Finally, associations between synchrony and PES cannot establish causation. Higher split-half reliability would indicate greater internal consistency under the chosen conditions, but would not by itself establish greater validity or stability across separate recording sessions.

Timeline

Table 1. Proposed Research Timeline — retained from the submission.

Work package	Main activities	Indicative timing
1. Finalise research protocol	Agree hypotheses, screening criteria and analysis settings.	Week 1
2. Ethics review	Submit the ethics application and address any requested revisions.	Weeks 2–3
3. Behavioural screening	Apply agreed exclusions to existing trial records.	Weeks 4–5
4. EEG preprocessing	Extend the reviewed pipeline and record usable trials.	Weeks 5–7
5. Calculate measures	Calculate PES, Δ ERN and theta synchrony.	Weeks 7–9
6. Statistical analysis	Test H1–H3 and conduct H4a reliability and H4b robustness analyses.	Weeks 10–12
7. Write-up	Complete results, discussion and limitations.	Weeks 13–14
8. Revision and submission	Proofread, check formatting and references.	Week 15

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